

# You-Zhe Xie (謝侑哲)

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## Education

📄 Transcript | 📖 Google Scholar | 🔗 linkedin | 🐙 github | 🐦 Twitter

**National Yang Ming Chiao Tung University**, BS in Computer Science

09.2023 – 06.2027

- Ranking: **1/197 (0.51%)**, GPA: 4.29/4.30
- **Language**: English (TOEIC 810), Chinese (Native)

## Experience

**Research Intern**, Alaya Lab, Shanda AI Research (Tokyo)

12.2025 - Now

- Research in **video generation models' understanding of causality** (YoCausal). Built the first benchmark for causal cognition in video generation models, extensible with arbitrary real video at zero cost. 1 first-author paper in submission to ICLR 2027.
- Building a benchmark for **extracting game mechanics from gameplay videos with MLLMs**.

**Research Assistant**, CPLab, NYCU (Advisor: Prof. Yu-Lun Liu)

06.2024 - Now

- Research in **3DGS poisoning attack** (StealthAttack). Stealthily inject illusory objects into a target view of 3DGS, achieving **SOTA** in this task. Published **1 ICCV paper**; delivered an **oral presentation** and won the **Best Poster Award** at an ICCV workshop.
- Research in **3D cinemagraph generation** (OuroWorld). Animate any static 3D scene into a seamlessly looping 3D cinemagraph. 1 first-author paper in submission to ICLR 2027.
- Research in **MLLM-guided physical parameter estimation from video** (TwinReason). 1 paper in submission to WACV 2027.

**Research Intern**, AIIULab, Academia Sinica (Advisor: Dr. Jun-Cheng Chen)

07.2025 - Now

- Research in **4D Gaussian ray tracing for physics-based, fine-grained camera effect data generation** (4DGRT). Achieve **highest data generation speed** and comparable quality to NeRF-based method.
- Published **1 NeurIPS workshop paper** and delivered an **oral presentation**.

**Consultant**, WorldQuant

02.2025 - Now

- Design strategy for **quantitative trading** to allocate investment funds. The best is to achieve **average 13.2% returns per year**.

**Research Assistant**, Department of Management Science, NYCU

02.2025 - 06.2026

- Research in the Lotka-Volterra model based on biological competition for sales forecasting. Analyze competitive relationships between different products.

**Mentor**, Google Developer Student Club, NYCU

09.2024 - 06.2025

- Taught courses on **AI and computer vision** and advised projects about **font style transfer & weakly supervised semantic segmentation**.

## Publications

**YoCausal: How Far is Video Generation from World Model? A Causality Perspective** 📄🌐🌐

03.2026

**You-Zhe Xie**, Yu-Hsuan Li, Jie-Ying Lee, Kaipeng Zhang, Yu-Lun Liu, Zhixiang Wang

ICLR 2027 (Under Submission)

**OuroWorld: Bringing Any 3D World Alive as Diverse, Endlessly Looping 3D Cinemagraphs** 🌐🌐

09.2026

**You-Zhe Xie**, Yu-Hsuan Li, Ting-Wei Chou, Zhixiang Wang, Yu-Lun Liu

ICLR 2027 (Under Submission)

**StealthAttack: Robust 3D Gaussian Splatting Poisoning via Density-Guided Illusions** 📄🌐🌐

03.2025

Bo-Hsu Ke, **You-Zhe Xie (2nd)**, Yu-Lun Liu, Wei-Chen Chiu

ICCV 2025 (Poster)

ICCV 2025 Responsible Imaging Workshop (★ Oral, 🌟 Best Poster)

**Every Camera Effect, Every Time, All at Once: 4D Gaussian Ray Tracing for Physics-based**

08.2025

**Camera Effect Data Generation** 📄🌐🌐

Yi-Ruei Liu, **You-Zhe Xie (co-1st)**, Yu-Hsiang Hsu, I-Sheng Fang, Yu-Lun Liu, Jun-Cheng Chen

NeurIPS 2025 SpaVLE Workshop (★ Oral)

**TGPED: Task-Generalized Procedural Error Detection and Explanation using Large Language**

09.2026

**Models with Procedural Grounding**

Siangyu Yang, S-Wei Chen, **You-Zhe Xie (3rd)**, Ho-En Tsai, Yu-Chi Lai, Kai-En Peng, YouChen Lai, Yu-Lun Liu, Chun-Yi Lee, Shih-Po Lee, Shao-Yuan Lo

ICLR 2027 (Under Submission)

**TwinReason: Cross-Engine MLLM-Guided Inverse Physics from Monocular Video**

03.2026

Chih-Hai Su, Fu-En Wang, Nan Qiao, Yuyin Sun, **You-Zhe Xie (5th)**, Yang-Sen Lin, Jie-Ying Lee, Yi-Ruei Liu,

Roberto Martín-Martín, Yu-Lun Liu, Tsung-Wei Ke, Min Sun, Cheng-Hao Kuo

WACV 2027 (Under Submission)

## Projects

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- ShellBench: A Parametric Shell Game Benchmark for Memory-Aware Manipulation** 🌐🔗 06.2026
- ShellBench is a parametric **Isaac Sim** shell-game benchmark that measures the **long-term memory** capacity of visuomotor manipulation policies by forcing them to track a ball hidden under shuffled identical cups. Cup count, shuffle count and speed are tunable, yielding memory-capacity curves that quantify each model's **memory decay**.
- Designing Fun: LLM-based Game Level Generation via Agent Gameplay Feedback** 🌐🔗 06.2026
- A two-stage SFT + **GRPO** framework that trains an LLM to generate fun MiniGrid levels by optimizing three gameplay-grounded rewards: regret, interaction, and diversity. Outperforms vanilla and SFT-only baselines on all metrics.
- MATCHA: Match with Agent** 🔗 04.2026
- Built at the 2026 YTP Hackathon. MATCHA is a **multi-agent** system where personal AI **Agents socialize** on users' behalf, using **RAG** to actively match them with government career resources and peer experiences.
- Desktop Organizer** 🌐🔗 12.2025
- Built an intuitive desktop organizer based on **user research** to ensure ease of use. Integrated **LLM** for automatic file management, designing a simple **React** interface that makes managing desktops effortless.
- Rewinded Video Revision** 12.2025
- Revise the background artifact of videos with rewind effect. Solve the problem by 3 stages: foreground **segmentation**, unreasonable **region detection**, **tuning-free video revision**. This project later **inspired YoCausal**.
- Real-Time DSTNet** 🌐🔗 08.2024
- Improved **video deblurring** model DSTNet to **real-time inference** (37 fps), **sped up to 400%**, refactored from offline to online.
- Anime Helper** 🌐🔗 12.2024
- Anime info. website with **React.js**, **Express.js**, **PostgreSQL**. I was responsible for the **backend** engineering.
- Paper Searching Bot with LLM** 🌐🔗📺 06.2024
- Built paper searching discord bot with **RAG** on LLM. Optimized result by prompt engineering. Collected data via crawler. Achieved **96.6% content relativity**.
- Deep Learning Projects** 🌐🔗 08.2024
- 6 sub-projects with technique including: CNN, UNet, ResNet, VAE, Transformer, Diffusion etc.
- Berkeley Computer Vision Projects** 🌐 08.2024
- 5 sub-projects include image alignment, hybrid, warping, stitching, auto stitching.

## Honors

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**First Place (特優), NYCU Undergraduate Project Competition** : Awarded for YoCausal.

**Academic Excellence Award (書卷獎) \* 5** : Top 5% in the class.

**Core Fundamental Course Award (核心基礎課程獎) \* 4** : Top 5% in specific courses.

- Algorithm, Discrete Mathematics, Digital Circuit Design, Computer Organization